Sauti Tri-Sense

Community-Led Early Warning for Population & Workforce Health

Future Health Challenge 2026 – To Sense is to Predict
Primary Challenge: Population and Workforce Health Early Warning
Track: Population Health | Team: Abdulkadir Ahmed – FCT Abuja, Nigeria

1. Problem

Nigeria's health early warning is reactive. Facility-based IDSR reports outbreaks 2-3 weeks late. 40% rural water points have intermittent chemical contamination (fluoride, heavy metals) untested. Informal workforce (paint, welding, markets) faces heat & chemical risks with no sensing. Existing tools require internet, English, expensive labs. In Abuja/Abaji & Kano, diarrhoeal clusters are reported only after hospitalization.

2. Solution – To Sense is to Predict

Sauti Tri-Sense (Sauti = Voice) turns community voices + phone camera into early warning system. 3 Modules:

Module 1 – Sauti Community: Volunteers log 30-sec voice note in Hausa/English via offline-first Android app. On-device Whisper-Hausa STT → LLM symptom extraction → geo-tagged event. Syncs when online.

Module 2 – Sauti ChemSafe: $0.20 strip + phone camera colorimetric reader → pH/conductivity vs WHO limit. Maps hotspot, triggers safe-water advisory.

Module 3 – Sauti Wellbeing: Hausa/English voice chatbot sustains handwashing, tracks referral completion to PHC, screens PHQ-2 for anxiety, closes screening-to-care loop.

3. Technology Type (As per form)

■ AI agent – Wellbeing Companion voice agent (Hausa/English motivational interviewing)

■ Machine learning – Fine-tuned Whisper-small-Hausa, Isolation Forest anomaly detection, entity extraction

■ Software application – React Native offline-first + FastAPI + PostgreSQL/TimescaleDB + LGA Dashboard

■ Analytics – Baseline vs anomaly time-series, hotspot heatmaps, workforce risk index

■ Sensor – Phone camera as sensor for colorimetric test, future low-cost conductivity sensor

■ Connected device – Phone as connected device for SMS alerts & sync

4. How It Works – Architecture

Volunteer (Tap) → Voice Hausa → On-device STT → Extract symptoms/count → Local PouchDB → When online → FastAPI → TimescaleDB → Anomaly detection (5x baseline in 48h) → Dashboard alert + SMS to

DSO/PHC.

Parallel: Water sample → Strip → Camera → OpenCV color correction → pH estimate → If fail → Map hotspot → Community advisory.

Closing Loop: Alert → Wellbeing bot calls household → Nudge PHC visit → Confirm attendance → Analytics update.

5. Responsible AI & Privacy

- On-device processing: Voice stays on phone unless consent.
- Federated learning for Hausa dialects.
- Explainable alerts: 'Flagged because 15 cases vs 3 baseline in Ward X'.
- Consent, opt-out, no child data.
- Bias mitigation: Tested on Hausa, Yoruba, Pidgin.

6. Innovation & Differentiation

Unlike LIMS/lab solutions, we sense what community already senses. No new hardware for MVP. Unlike USSD surveys, we use natural voice + camera. Combines 3 senses: Community voice (population), Water sense (chemical), Wellbeing sense (behaviour/mental).

7. Impact – Abuja Pilot

Pilot: 10 wards, 50 volunteers, 6 months. KPIs: Detect outbreak 7-14 days earlier, reduce water-borne cases 25%, increase screening-to-care completion from 30% to 65%, 10k households reached. Scalable to ECOWAS. Workforce health: Monitor heat + chemical exposure for informal workers.

8. Team – Why Us

Abdulkadir Ahmed, HND Chemical Engineering (2015) Federal Polytechnic Bida, Branch Manager Manglam Tours (market intelligence, lead gen), Ex-Production Manager Zancotex Paint (formulation, concentration, conductivity, HAZMAT, GLP). Customer Care & Complaint Management Certified (Oct 2026). Native Hausa, Fluent Yoruba, Based FCT Abuja. 7+ years field ops & community structures. Linked referees in Kano & Abuja.

9. Prototype Link & Next Steps

Prototype: https://github.com/AbdulkadirAhmed/Sauti-Tri-Sense (MVP in dev). Figma: Mobile App + Dashboard. Live Demo by 18 Nov 2026: https://lovable.dev/projects/sauti-tri-sense-demo
Roadmap: Nov – MVP build, Dec – Pilot with 2 PHCs Abaji, Jan – LGA integration, Feb – Publish Hausa dataset.

10. Alignment with Challenge Statements

Primary: Population and Workforce Health Early Warning – 100% fit.
Secondary: Precision Prevention & Screening, Closing Screening-to-Care Loop, Climate-Resilient Worker

Health, Sustaining Healthy Behaviors, Earlier Safer Mental-Wellbeing Support – covered by Modules 2 & 3.